FPT SCHOOL OF BUSINESS & TECHNOLOGY

MASTER OF SOFTWARE ENGINEERING

SYLLABUS

Course Name:	**Big Data**
Course Code:	**BDA501**
No of credits:	**03**
Time Allocation:	Study hours (150h) - Contact hours (45h) - Final assessment (3h) - Self-study (102h)
Prerequisite:	PPR501

I. DESCRIPTION

Big Data Analytics addresses the collection, storage, processing and analysis of data at scales or rates that require distributed and parallel computing. The course introduces modern Big Data Analytics architecture from data ingestion and distributed/object storage to batch and stream processing, interactive analytics, NoSQL serving, machine learning and cloud/cluster deployment. Hadoop/HDFS and MapReduce are retained as foundations for distributed processing, while Apache Spark with Python, DataFrames/Spark SQL and Spark ML are the main implementation platform. Students implement a MapReduce workflow, build Spark data pipelines, work with representative NoSQL technologies, apply machine learning, and design scalable Big Data solutions.

II. MAIN OBJECTIVES

Upon successful completion of this course, students will be able to:

1. Understand Big Data frameworks, including Hadoop, HDFS, MapReduce, Pig, and Hive.
2. Develop and optimize distributed applications using MapReduce and Spark with Java/Python.
3. Analyze data and perform parallel programming using Big Data tools like Spark, and Pig, Hive.
4. Apply machine learning algorithms and integrate Spark with NoSQL databases and other Big Data technologies.
5. Deploy and scale Big Data applications on clusters and in Cloud environments.

Mapping CLOs to PLOs:

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	X											
CLO2		X										
CLO3		X	X	X								
CLO4		X	X	X				X				
CLO5			X	X				X				

III. TEACHING METHODS

To achieve the best course objectives, teaching methods and activities are used spontaneously, including:

- Teaching theory
- Group activities
- Group presentations

In addition, during the learning process, faculty can use different methods to achieve the teaching goals in the best way.

IV. STUDENT'S TASK

- Students are responsible for doing all presentations and projects/problems assigned by instructor and submitting them on time.
- Promptly access to the FSB AP for up-to-date course information.

V. TEACHING & LEARNING MATERIALS

Instructors' presentation:

Chapters' Presentations

Textbooks:

1. Arshdeep Bahga & Vijay Madisetti, Big Data Analytics: A Hands-On Approach, 2019, ISBN 978-1-949978-00-1. Selected topics: Big Data stack and architecture patterns; NoSQL; data acquisition; HDFS; MapReduce; Spark; Spark SQL; serving databases; Spark MLlib.
2. Jure Leskovec, Anand Rajaraman & Jeffrey D. Ullman, Mining of Massive Datasets, Cambridge University Press, 3rd Ed., 2020. Selected topics: MapReduce and the new software stack; data streams; clustering; recommendation systems; large-scale machine learning.
3. Apache Spark Documentation (https://spark.apache.org/docs/latest/)
4. Apache Hadoop Documentation (https://hadoop.apache.org/docs/current/)

Other materials:

- Spark SQL, DataFrames and Datasets Guide
- Spark Structured Streaming Guide

- Spark MLlib Guide
- Spark Cluster Mode Overview
- MongoDB Connector for Spark
- Apache Cassandra Documentation
- Apache Kafka Documentation
- Python 3 Documentation

Required self-study topics:

Pig/Pig Latin; Hive/HiveQL and Hive concepts; selected YARN administration; Sqoop; Flume; Impala/Oozie; HDP/CDH setup concepts; legacy Spark Streaming/DStreams; legacy RDD-based MLlib API.

VI. SCHEDULE

Session	Module	CLO
1	Trial Exam, Fundamental Concepts & Project Handout: Modern Big Data Analytics Architecture, MapReduce Foundations & Final Project Launch • Big Data characteristics, analytics flow and Big Data stack. • Modern architecture: sources/ingestion; distributed or object storage; processing; SQL/analytics; machine learning; NoSQL serving; visualization. • HDFS foundations: blocks, replication, fault tolerance and data locality. • MapReduce: map, key-value output, shuffle/sort/grouping, reduce and combiner concepts. • Batch versus streaming; sharding, CAP and scalable architecture patterns. • Final project (40%) requirements and project scope.	CLO1, CLO2, CLO5
2	Individual Exercise 1 (10%): Python MapReduce Workflow • Environment verification for Python, Hadoop/HDFS or the course-approved reproducible environment. • Python mapper and reducer for a meaningful log/event/transaction aggregation. • Execution with Hadoop Streaming or equivalent course environment. • Map -> shuffle/group -> reduce explanation, combiner validity and baseline result validation. • Submission of source code, execution evidence, README and technical explanation.	CLO1, CLO2
3	Apache Spark Architecture, DataFrames & Spark SQL • Spark architecture: driver, executors, jobs, stages, tasks, partitions, lineage and fault recovery. • RDD concepts as a bridge from MapReduce: transformations/actions, lazy evaluation and shuffle. • PySpark DataFrames: schemas, filtering, projection, joins, aggregation and built-in functions. • Spark SQL, temporary views and equivalent SQL/DataFrame expressions. • CSV/JSON/Parquet, partitioning and query-plan concepts.	CLO2, CLO3
4	Individual Exercise 2 (10%): PySpark Data Engineering & Spark SQL • Build a reproducible DataFrame pipeline with schema, cleaning, transformations, join/enrichment and aggregation. • Answer analytical questions using DataFrame API and Spark SQL. • Write curated Parquet output and justify partitioning/data-layout decisions. • Inspect query-plan evidence and identify shuffle/Exchange behavior. • Submit code/notebook, README, outputs and technical explanation.	CLO2, CLO3

Session	Module	CLO
5	Team Presentation 1 (10%): Final Project Proposal; NoSQL & Spark ML Foundations • Final project proposal: problem, Big Data justification, dataset, architecture, expected analytics/ML outcome, evaluation plan and risks. • NoSQL models: key-value, document, wide-column and graph; CAP and access-pattern considerations. • MongoDB and Cassandra as representative serving databases. • Spark ML concepts: DataFrame-based Pipeline, transformers, estimators, feature vectors and evaluation. • Project architecture refinement for the Session 10 final assessment.	CLO1, CLO4, CLO5
6	Individual Exercise 3 (10%): Spark ML Pipeline & NoSQL Serving • Build one Spark ML pipeline for clustering, classification or regression. • Feature preparation, model fitting/prediction and appropriate evaluation. • Persist predictions, cluster assignments or scored results to MongoDB or another approved NoSQL store. • Read back and verify persisted output; explain the NoSQL data model and access pattern. • Submit code/notebook, README, evaluation evidence, NoSQL evidence and technical explanation.	CLO2, CLO4
7	Data Ingestion, Streaming & Cloud/Cluster Deployment • Kafka/pub-sub concepts and ingestion patterns. • Structured Streaming: DataFrame-based stream processing, source/sink, checkpointing and event-time/state awareness. • Spark deployment architecture: Standalone, YARN, Kubernetes and managed/cloud Spark concepts. • Object storage, compute/storage separation, scalable serving and deployment architecture. • Operational concerns: partitioning, fault tolerance, observability, security and cost/performance trade-offs.	CLO4, CLO5
8	Team Presentation 2 (10%): Final Project Progress Review • Working ingestion/data-processing evidence and current Spark SQL pipeline. • Preliminary analytics/ML and NoSQL evidence. • Updated architecture showing implemented versus planned components. • Current technical risks, performance issues and corrective plan. • Plan to complete the final report, implementation and demonstration for Session 10.	CLO1-CLO5
9	Integrated Big Data Analytics & Final Project Preparation • End-to-end integration: from ingestion/storage to Spark SQL, ML, NoSQL serving/output. • Performance and scalability: partitions, shuffle, file layout, caching/broadcast choices and query plans. • Testing and validation of data-processing and analytics results. • Cloud/cluster deployment design, streaming consideration, reproducibility and technical limitations. • Final project completeness check against the required technical deliverables.	CLO1-CLO5
10	FINAL ASSESSMENT (40%): Final Project Report, Presentation, Demo & Q&A • Present the Big Data Analytics architecture and implemented solution. • Demonstrate the MapReduce component, Spark DataFrame/Spark SQL pipeline, machine learning and NoSQL serving evidence. • Explain streaming and cloud/cluster deployment design where applicable. • Answer technical questions on implementation decisions, distributed processing, scalability and course learning outcomes. • Submit the final report and complete project package.	CLO1-CLO5

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VII. ASSESSMENT

Assessment type	Symbol	Weight
ON-GOING		60%
1. Participation	P	10%
2. Individual (3 exercises)	I	30%
3. Team (2 theory presentation)	T	20%
4. Quiz	Q	
FINAL ASSESSMENT (Assignment)	FE	40%

Completion Criteria: Average>=5 & Final assessment >=4 (Score:10)

Detailed assessment in the Appendix

APPENDIX: DETAILED ASSESSMENT

Assessment Category	Assessment Type	Weight	Minimum value to meet completion criteria	Duration	Learning outcomes	Number of questions	Scope of knowledge and skill of question	How
Participation	On-going	10	N/A	Throughout all sessions	CLO1-CLO5			The P score will be evaluated by the lecturer based on information provided by the class assistant, with the following criteria: - Students' attendance across all sessions; - Level of participation and contributions during in-class activities; - Constructive feedback provided to improve training quality via evaluation forms
Exercise	On-going	30	N/A	Sessions 2, 4, 6	CLO1-CLO5	Depend on exercise	The content of the topics under the module	Evaluation is based on the specific technical requirements and rubrics of Individual Exercises 1, 2 and 3. Each exercise contributes 10% to the course grade.
Theory presentation	On-going	20	N/A	Sessions 5 and 8	CLO1-CLO5	2	Studied chapters knowledge and project skills	Evaluation is based on the specific requirements of the presentation. Presentation 1 covers the final-project proposal; Presentation 2 covers project progress and technical evidence. Each presentation contributes 10% to the course grade.
Assignment	Final	40	4	Session 10	CLO1-CLO5	1	Summary and application of the content of all topics	Evaluation based on the following criteria: - Accuracy and Correctness: Ensure the solutions are correct and error-free. - Creativity and Problem-Solving Ability: Demonstrate innovative solutions and critical thinking. - Completeness of the Product: Provide all required components and supporting documentation. - Presentation and Expression Skills: Present the assignment clearly with correct terminology. - Application of Theoretical Knowledge: Apply theory effectively to practical problems. - Timeliness and Adherence to Requirements: Submit on time and meet all assignment requirements.

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